Partial differential equations: advection and characteristics

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Where we are

- Weeks 4 to 7: ordinary differential equations. One independent variable, usually time; initial value problems and boundary value problems.
- Weeks 8 to 9: linear algebra, because the grid method turns a boundary value problem into $\mathbf{A}\mathbf{x} = \mathbf{b}$.
- From today: partial differential equations. Unknowns that depend on space *and* time, and the methods that fluid, plasma and electromagnetic simulations are built on. Project 2 will use them.

Today: the simplest PDE there is, the one idea that explains how its solutions move, and the first scheme that gets it right.

Part I · The advection equation

A profile carried along at constant speed

What a PDE is

An equation for a function of several variables, involving its partial derivatives. The ones physics runs on:

diffusion	$\partial_t u = D\,\partial_x^2 u$	heat, particles spreading
wave	$\partial_t^2 u = c^2\,\partial_x^2 u$	strings, sound, light
Schrödinger	$i\hbar\,\partial_t \psi = -\frac{\hbar^2}{2m}\partial_x^2 \psi + V\psi$	quantum mechanics in time
Poisson	$\partial_x^2 \phi + \partial_y^2 \phi = -\rho/\varepsilon_0$	electrostatics, gravity
Euler	$\partial_t \rho + \partial_x(\rho v) = 0$, and two more	fluids and gases

- Same two kinds as for ODEs. Diffusion, wave and Schrödinger are *initial value problems*: given u at $t = 0$, march forward. Poisson is a *boundary value problem*: ϕ on the boundary, solve for the inside.
- Same plan as BVP in ODEs: put space on a grid, replace derivatives by differences, and the PDE becomes a large system of ODEs, or a large linear system.

The simplest PDE

Something is carried along at a constant speed v, without changing shape. Dye in a pipe, a pulse on a conveyor belt, a density of cars on a road where everyone drives at v. Its density $u(x, t)$ obeys

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial x} = 0.$$

- First order in t and in x, linear, with one parameter. Called the *advection equation*: advection is transport by a flow.
- An initial value problem: give $u(x, 0) = u_0(x)$ and ask for $u(x, t)$ later.

Everything in fluid dynamics has a term of this form in it. If a method cannot do this equation, it cannot do fluids.

Its exact solution

Guess that the profile just moves: $u(x,t) = f(x - vt)$ for some function f. Check it with the chain rule, with f' the derivative of f with respect to its argument:

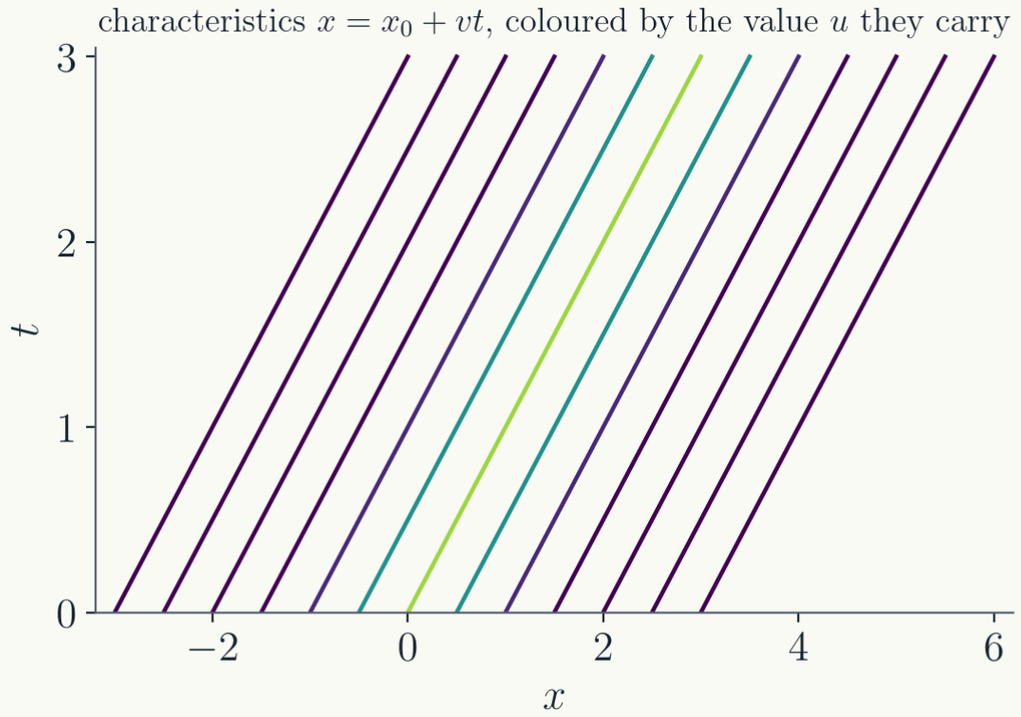
$$\frac{\partial u}{\partial t} = -v\, f'(x - vt), \qquad \frac{\partial u}{\partial x} = f'(x - vt), \qquad \frac{\partial u}{\partial t} + v\,\frac{\partial u}{\partial x} = 0.$$

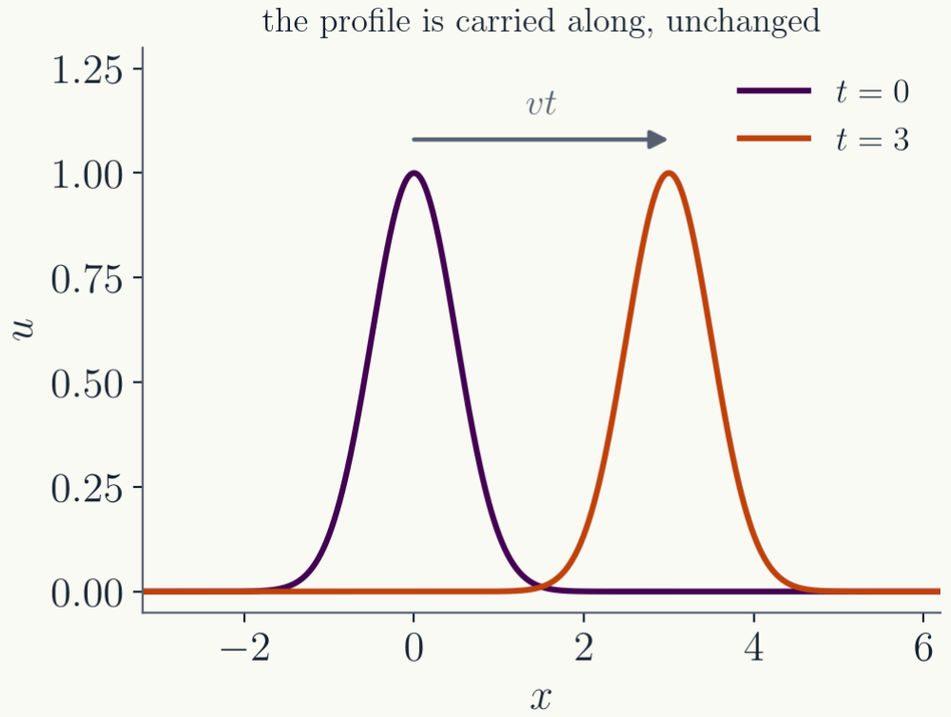
- It works for *every* f. The initial condition picks it: $u(x,0) = f(x)$, so $f = u_0$ and

$$u(x,t) = u_0(x - vt)$$

- The profile at time t is the initial profile shifted by vt. No spreading, no decay, no change of shape. A kink stays a kink.

Characteristics





Follow a point that moves with the flow, $x(t) = x_0 + vt$. The value of u seen from it changes at the rate

$$\frac{d}{dt}\,u\big(x(t),t\big) = \frac{\partial u}{\partial t} + \frac{dx}{dt}\frac{\partial u}{\partial x} = \frac{\partial u}{\partial t} + v\,\frac{\partial u}{\partial x} = 0.$$

u is constant along these lines. They are the characteristics of the equation, and the PDE says: nothing happens along them.

What the lines tell you

- The value at (x, t) depends on the initial data at *one* point, $x - vt$. Nothing else matters.
- Information travels at speed v, no faster and no slower. A change made at x_0 at $t = 0$ is felt at x exactly at $t = (x - x_0)/v$.
- On a finite domain $[0, L]$ with $v > 0$, the characteristics enter at $x = 0$ and leave at $x = L$. So a boundary condition is needed at the *inflow* end only. Prescribing u at the outflow end would contradict what the characteristic brings there.
- Sharp features are preserved. The equation does not smooth anything; a numerical method that does is adding something of its own.

Each of these will come back when we build a scheme: where to take information from, how far it can travel in one step, where boundary conditions belong.

Part II · Characteristics beyond the toy

Four changes to the equation, one idea

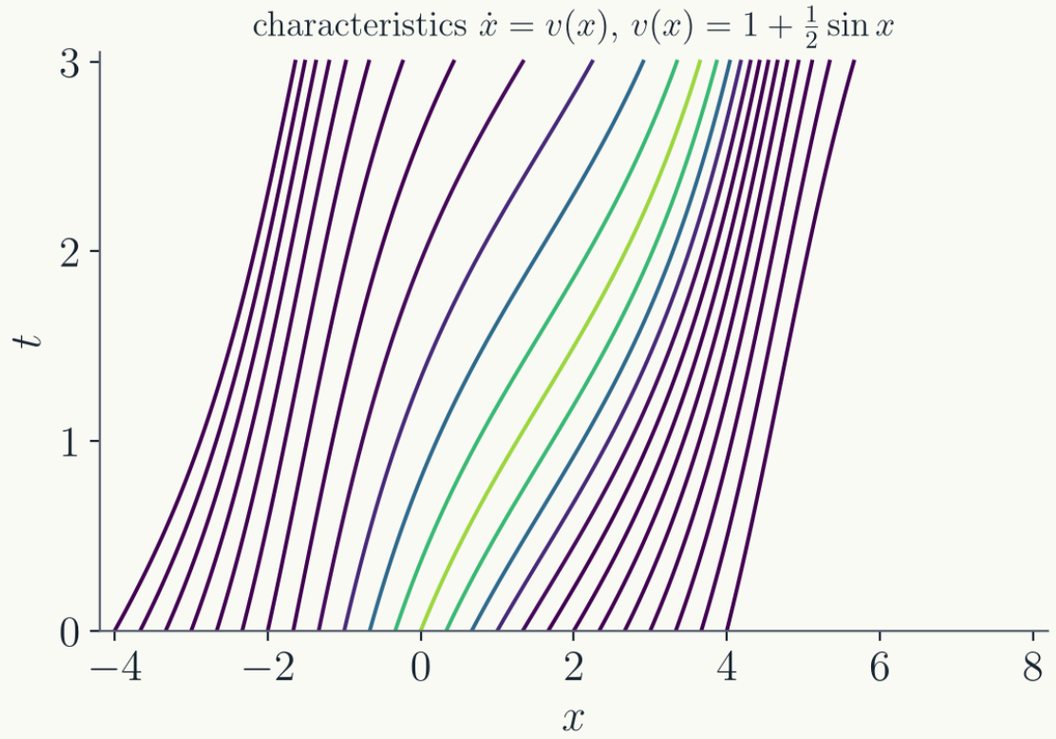
Variable speed

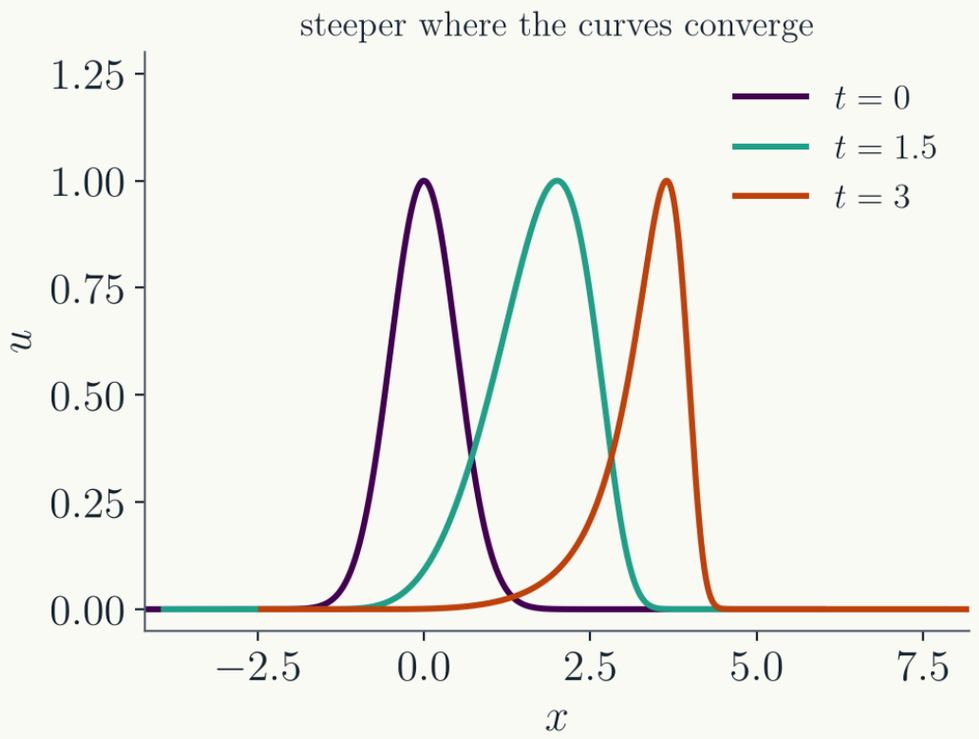
Let the speed depend on position: $\partial_t u + v(x)\,\partial_x u = 0$. Follow a point moving with the local flow, $dx/dt = v(x)$. The chain rule again:

$$\frac{d}{dt}\,u\big(x(t),t\big) = \frac{\partial u}{\partial t} + v(x)\,\frac{\partial u}{\partial x} = 0.$$

- u is still constant along each curve. The curves are now solutions of the ODE $dx/dt = v(x)$, one through each starting point x_0: curved, not straight.
- The solution: $u(x,t) = u_0(x_0)$, where x_0 is the starting point of the characteristic that reaches (x,t).

Where the curves converge, the profile steepens





- $v(x) = 1 + \frac{1}{2}\sin x$. Where v decreases with x the front of the pulse slows before the back does, the characteristics bunch, and the profile compresses: its steepest slope doubles, from 1.21 to 2.42, by $t = 3$.
- The values are unchanged; only their spacing is. The maximum is still 1.

A source term

Let something happen to u along the way, a decay, a chemical reaction, a force: $\partial_t u + v\,\partial_x u = s(x,t,u)$. Along $dx/dt = v$ the same chain rule gives

$$\frac{d}{dt}\,u\big(x(t),t\big) = s\big(x(t),t,u\big).$$

- u is no longer constant along the characteristic, but it obeys an *ordinary* differential equation there, with $x(t)$ known.
- Example: $s = -u/\tau$. Along each line u decays as $e^{-t/\tau}$, so $u(x,t) = u_0(x - vt)\,e^{-t/\tau}$.

A first-order PDE is a family of ODEs, one along each characteristic. You already know how to solve ODEs.

The recipe

For any equation of the form

$$\frac{\partial u}{\partial t} + a(x,t,u)\,\frac{\partial u}{\partial x} = s(x,t,u),$$

solve, from every starting point x_0 with $u = u_0(x_0)$, the pair of ODEs

$$\frac{dx}{dt} = a(x,t,u), \qquad \frac{du}{dt} = s(x,t,u).$$

- The first says where the characteristic goes; the second says what u does along it. They are coupled when a depends on u.
- This is the *method of characteristics*. It turns one PDE into many ODEs, and it works as long as the characteristics do not cross.

Nonlinear speed: Burgers' equation

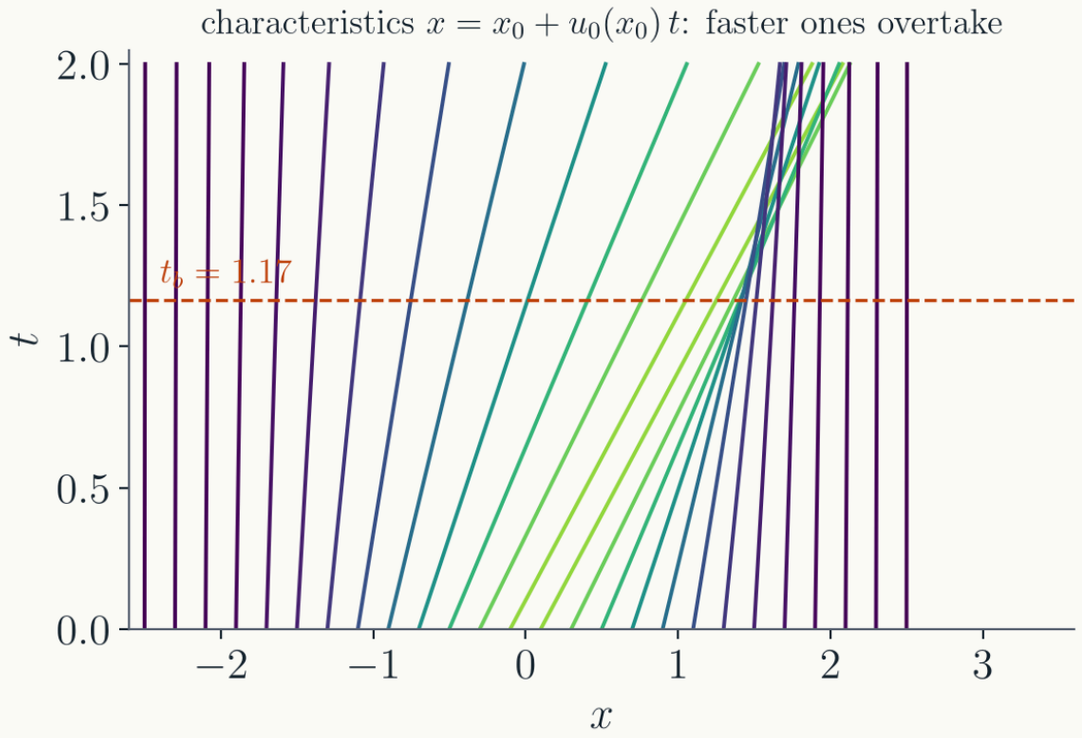
Let the speed be the value itself, $a = u$. The equation

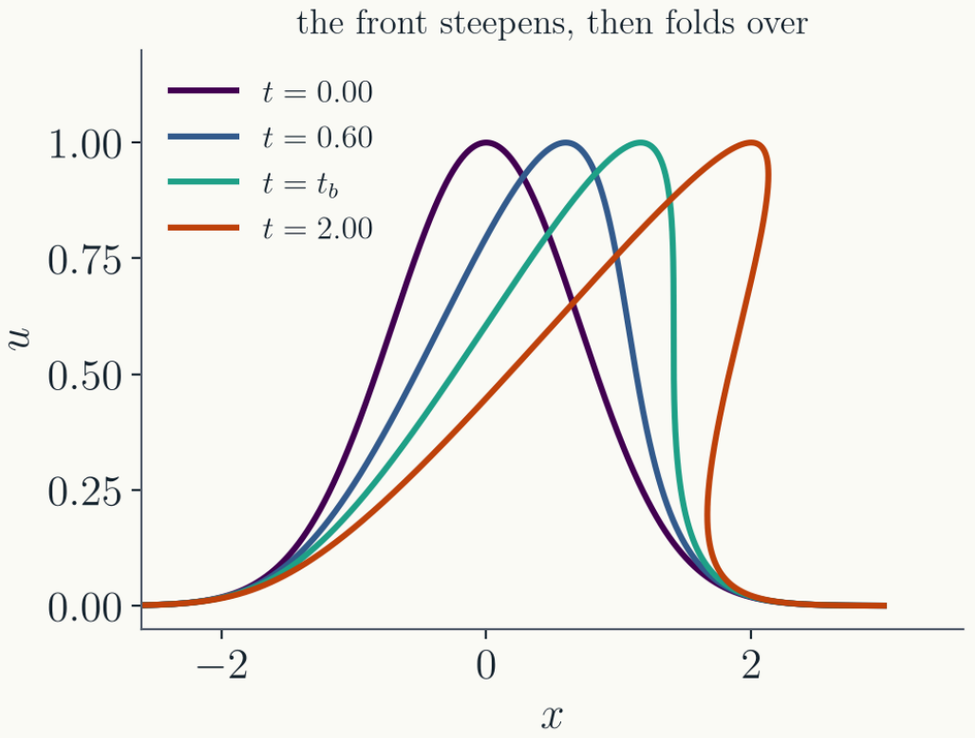
$$\frac{\partial u}{\partial t} + u\,\frac{\partial u}{\partial x} = 0$$

is the simplest model of a fluid: u is the velocity, and each bit of fluid keeps its velocity and moves with it. The recipe gives $dx/dt = u$ and $du/dt = 0$:

- u is constant along each characteristic, so each characteristic is a straight line, with slope set by the value it carries: $x = x_0 + u_0(x_0)\,t$.
- Where u_0 is large the lines are fast. Fast parts of the profile overtake slow parts.

They cross





- Two neighbouring characteristics from x_0 and $x_0 + dx_0$ meet when $x_0 + u_0 t = x_0 + dx_0 + (u_0 + u_0'\,dx_0)\,t$, that is at $t = -1/u_0'(x_0)$.
- The first crossing is the *breaking time*, $t_b = -1/\min u_0'$. For $u_0 = e^{-x^2}$, $\min u_0' = -0.858$ at $x_0 = 0.71$, so $t_b = 1.17$.

After the breaking time

- Past t_b the construction gives two or three values of u at one x. For a velocity field that is meaningless: the smooth solution has ended.
- Physics resolves it with a *shock*: a jump in u that moves so that the stuff is still conserved. Written as a conservation law, $\partial_t u + \partial_x \left(\tfrac{1}{2} u^2\right) = 0$, the equation still makes sense across a jump, and the jump between u_L and u_R travels at the average speed $(u_L + u_R)/2$.
- The characteristics run *into* the shock from both sides. Information is lost there, which is why the solution past t_b is not reversible.

The method of characteristics told us exactly when and where the smooth solution fails. A simulation of a gas must survive that moment; the conservation form is how.

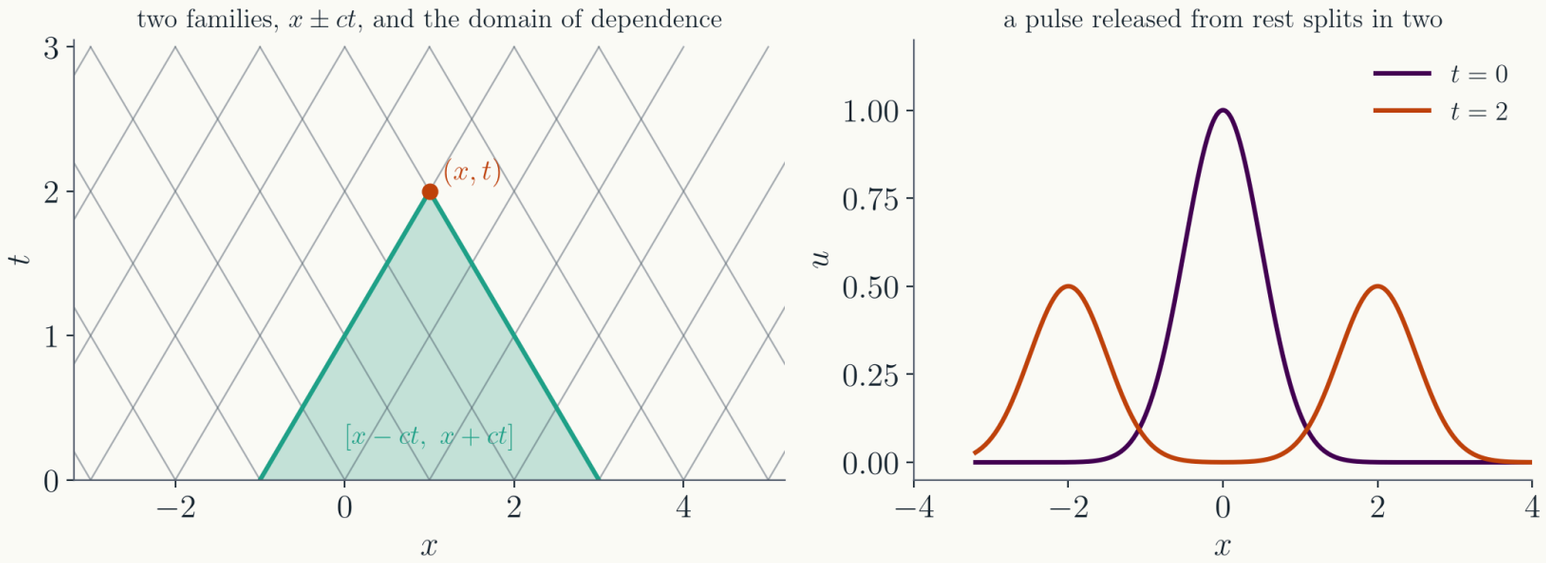
Two families: the wave equation

The wave equation is second order, but it factorizes into two first-order advection operators:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right) u = 0.$$

- Anything killed by either factor is a solution: $f(x - ct)$ moves right, $g(x + ct)$ moves left, and the general solution is $u = f(x - ct) + g(x + ct)$ (d'Alembert).
- Two families of characteristics, $x - ct = \text{const}$ and $x + ct = \text{const}$, each carrying its own piece of the solution.
- A pulse released from rest splits into two halves of height $\frac{1}{2} u_0$ moving apart.

The domain of dependence



- The value at (x, t) depends only on the initial data in $[x - ct,\ x + ct]$: the two characteristics through the point, traced back.
- For advection the interval was a single point. For a gas there are three families, at v and $v \pm c_s$, with c_s the sound speed.

Any numerical scheme has a domain of dependence too: the grid points it reads. It had better contain the physical one.

Part III · From characteristics to a scheme

Back to constant v, and onto a grid

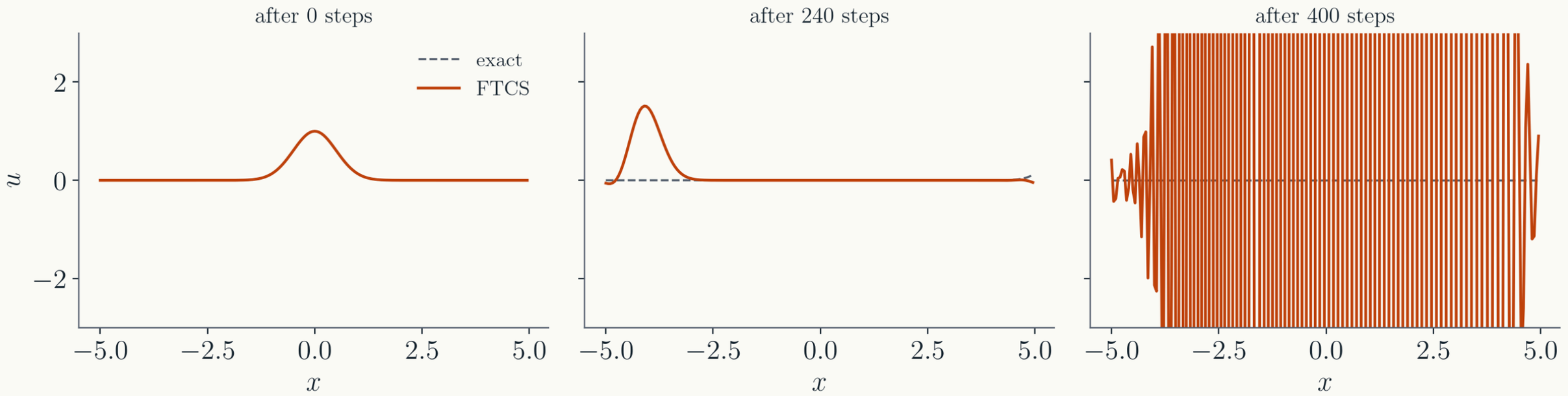
The grid

Points $x_j = j\,\Delta x$ and times $t_n = n\,\Delta t$; the unknowns are $u_j^n \approx u(x_j, t_n)$. Replace the derivatives by the differences of Lecture 12, centered in space and forward in time:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + v\,\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0 \qquad \Longrightarrow \qquad u_j^{n+1} = u_j^n - \frac{v\Delta t}{2\Delta x}\left(u_{j+1}^n - u_{j-1}^n\right).$$

- Each new value comes from three old ones. This is Euler's method applied to the system of ODEs $du_j/dt = -v\,(u_{j+1} - u_{j-1})/2\Delta x$, one per grid point. The scheme is called FTCS: forward time, centered space.
- The one number that matters is the combination $C = v\Delta t/\Delta x$: how many grid spacings the flow moves in one step.

FTCS fails

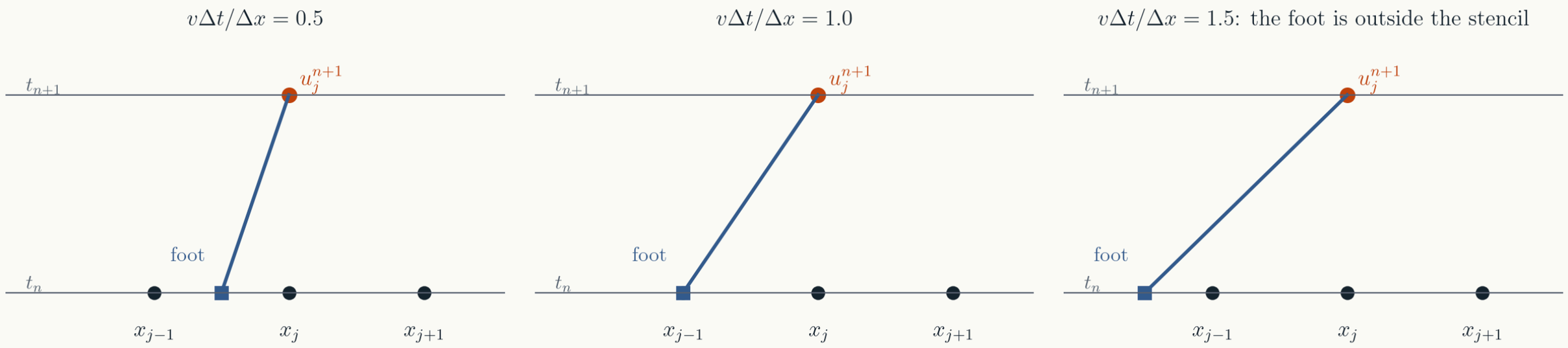


- $C = 0.5$, 200 grid points, periodic box. The peak grows to 1.5 after 240 steps and the box is filled with a grid-scale oscillation of amplitude 2×10^3 after 400.
- Smaller Δt does not save it; it only delays the blow-up. The scheme is unstable for every C. Next lecture shows why, by following a single wave on the grid.

Ask the characteristic

The exact solution says where the new value comes from: go down the characteristic for one step,

$$u(x_j, t_{n+1}) = u(x_j - v\Delta t,\ t_n).$$



- The foot of the characteristic, $x_j - v\Delta t$, is not a grid point. Read the old solution there by linear interpolation between the two grid points around it, x_{j-1} and x_j.

The upwind scheme

The foot lies a fraction $C = v\Delta t/\Delta x$ of the way from x_j back to x_{j-1}. Linear interpolation gives

$$u_j^{n+1} = (1 - C)\, u_j^n + C\, u_{j-1}^n.$$

- Rearranged, it is a difference scheme:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + v\,\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0.$$

  The space derivative is one-sided, taken from the side the flow comes from: *upwind*. For $v < 0$ use $u_{j+1} - u_j$ instead.

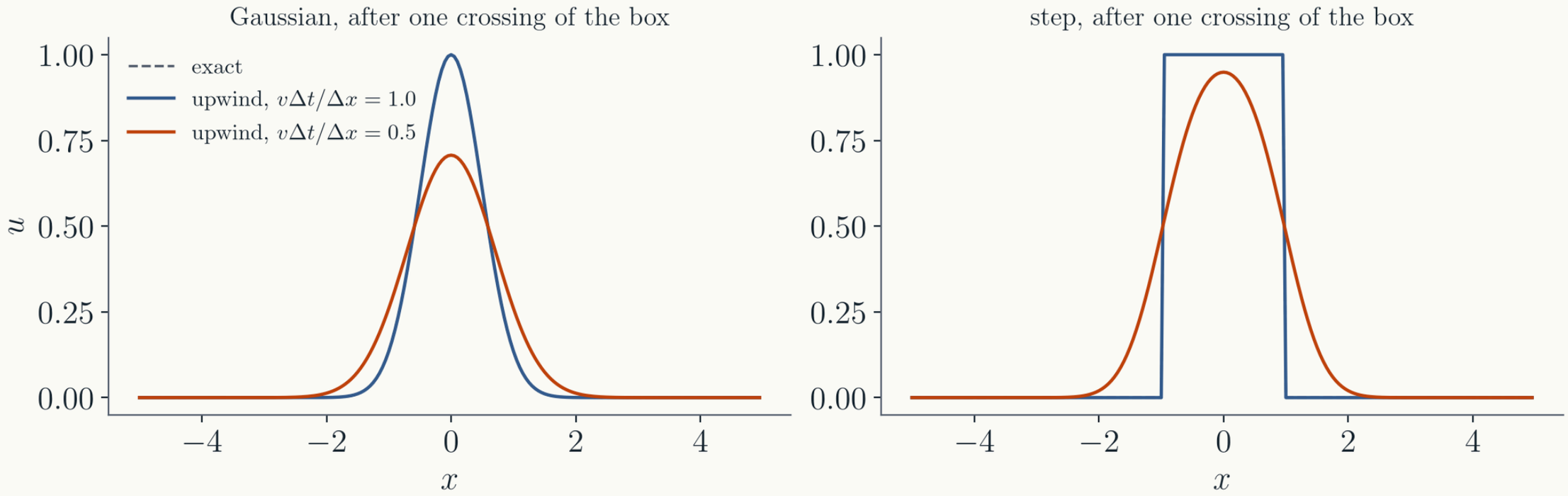
- Same cost as FTCS, and it only ever reads information from where the characteristic says it is.

When it is exact, and when it breaks

- $C = 1$: the foot lands exactly on x_{j-1}, and the scheme is $u_j^{n+1} = u_{j-1}^n$, a pure shift. The exact solution, to rounding.
- $C > 1$: the foot lies beyond x_{j-1}, outside the two points used. Interpolation has become extrapolation, with a negative weight $1 - C$, and the errors grow every step.
- $C < 1$: stable, but each step averages two neighbours, and averaging smooths. The profile smears a little every step.

$C = v\Delta t/\Delta x \leq 1$ is the CFL condition (Courant, Friedrichs, Lewy, 1928): in one step, the flow may not move farther than the scheme can see. The numerical domain of dependence must contain the physical one.

Upwind in action



- One crossing of the box. At $C = 1$ both profiles return exactly. At $C = 0.5$ the Gaussian peak has fallen to 0.71 and the step's corners are gone: the averaging of 400 steps.
- The smearing is the price of stability in this scheme. It looks like diffusion, and next lecture shows it *is* diffusion, with a coefficient set by Δx and C.

Summary

- **Advection:** $\partial_t u + v\,\partial_x u = 0$ carries the profile along, $u = u_0(x - vt)$.
- **Characteristics:** the curves $dx/dt = a$ along which the PDE becomes the ODE $du/dt = s$. Variable speed bends them, a source changes u along them, nonlinearity makes them cross: a shock, at $t_b = -1/\min u_0'$.
- **Domain of dependence:** where the value at (x, t) comes from. One point for advection, an interval for waves, three families for a gas.
- **Upwind:** read the old solution at the foot of the characteristic. Stable for $C = v\Delta t/\Delta x \le 1$, the CFL condition, exact at $C = 1$, smearing below it.

Next time: why FTCS blows up and upwind smears, by following one wave through a step. Then schemes that are second order: Lax, Lax-Wendroff, and leapfrog for the wave equation.